

Heart Disease Analysis

Phase 2: Requirement Analysis

Solution Requirement

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Functional Requirements

- Store and query the 4,500-record heart disease dataset in a relational database (MySQL).
- Run SQL analyses: prevalence rate, demographic breakdowns, lifestyle vs. disease correlation, high-risk profiling.
- Build an interactive Tableau dashboard with KPI cards, charts, and filters (Gender, Heart Disease, Smoking, Alcohol Drinking).
- Build a Tableau Story presenting a guided narrative: overview → demographics → lifestyle risk → medical risk → key insights.
- Publish the dashboard/story to Tableau Public for shareable, embeddable access.
- Embed the published dashboard into a lightweight Flask web application with a simple UI.
- Document the project (problem statement, methodology, insights, recommendations) and host source files on GitHub.

Non-Functional Requirements

- Usability: dashboard should be understandable to non-technical healthcare stakeholders.
- Performance: dashboard should load and filter within a few seconds for 4,500 rows.
- Availability: published dashboard and Flask app should be accessible via a public link.
- Maintainability: SQL scripts, dataset, and Tableau workbook version-controlled in GitHub.
- Portability: Tableau workbook must run on Tableau Public (no live DB connections; extract-based data only).
- Security: dataset contains no personally identifiable patient information.

Assumptions & Constraints

- Dataset is static (no real-time data ingestion required for this phase).
- Tableau Public does not support live database connections — all data sources must be extracts.
- Free-tier hosting (Render / PythonAnywhere) is sufficient for the Flask demo app.